

PERIORBITAL CELLULITIS – COMPREHENSIVE NCLEX OVERVIEW

DEFINITION

Periorbital Cellulitis (also called Preseptal Cellulitis) is an infection of the eyelid and surrounding skin tissues in front of the orbital septum (a connective tissue barrier that separates the front of the eye from the deeper orbital structures).

 It is a serious but usually non-vision-threatening condition if treated promptly.

DO NOT CONFUSE WITH: ORBITAL CELLULITIS

Periorbital Cellulitis	Orbital Cellulitis
Infection anterior to (in front of) the orbital septum	Infection posterior to (behind) the septum (within the orbit)
More common, less dangerous	Rare, but more severe & sight-threatening
No eye movement pain, no vision loss	Eye movement pain, proptosis (bulging eye), possible vision loss

CAUSES / RISK FACTORS

- Bacterial infection, often from:
 - Sinusitis (especially ethmoid sinus)
 - Eyelid trauma (insect bite, scratch, injury)
 - Chalazion or styne infection
 - Spread from upper respiratory tract infection
- Common organisms:
 - *Staphylococcus aureus*
 - *Streptococcus pneumoniae*

- Haemophilus influenzae (in unvaccinated children)

SIGNS & SYMPTOMS

- ✓ Eyelid swelling, redness, and tenderness
- ✓ Warmth over the eyelid
- ✓ Fever may be present
- ✓ No pain with eye movement
- ✓ No vision changes
- ✓ Eye movements and pupil responses are normal
- ✓ May have history of insect bite, sinus infection, or trauma

DIAGNOSIS

1. Physical Exam:
 - Distinguish from orbital cellulitis (which has deeper involvement)
2. CT scan of orbits with contrast (if uncertain or worsening symptoms)
 - Rules out orbital cellulitis or abscess
3. CBC (Complete Blood Count):
 - May show elevated white blood cell count (WBC) indicating infection
4. Blood cultures if febrile or toxic-appearing

TREATMENT

Severity	Treatment
Mild cases (outpatient)	Oral antibiotics (e.g., amoxicillin-clavulanate)
Moderate/severe cases or <1 year old	IV antibiotics (e.g., ceftriaxone, vancomycin) in hospital
Complications or no improvement	May need surgical drainage or ophthalmology referral



NURSING CONSIDERATIONS

- ✓ Monitor temperature and vital signs regularly
- ✓ Assess for progression to orbital cellulitis:
 - Pain with eye movement
 - Bulging eye (proptosis)
 - Double vision (diplopia)
 - Decreased visual acuity
- ✓ Administer antibiotics on time (oral or IV as prescribed)
- ✓ Apply warm compresses to the eyelid (if approved by provider)
- ✓ Encourage medication compliance at home to prevent complications
- ✓ Educate caregivers to return if symptoms worsen



NCLEX PRACTICE QUESTIONS

QUESTION 1

A 6-year-old child is brought in with eyelid swelling, redness, and warmth but no eye pain with movement and normal vision. What is the most likely diagnosis?

- A. Orbital cellulitis
- B. Allergic conjunctivitis
- C. Periorbital cellulitis
- D. Acute glaucoma

These are classic signs of **periorbital (preseptal) cellulitis**—infection in front of the orbital septum.

QUESTION 2

Which finding would cause the nurse to suspect **orbital cellulitis** instead of periorbital cellulitis?

- A. Mild eyelid redness
- B. No fever
- C. Pain with eye movement
- D. Normal extraocular movement

Pain with eye movement suggests deeper infection → possible orbital cellulitis.

QUESTION 3

Which teaching point is **most important** for parents of a child discharged with periorbital cellulitis?

- A. "Apply steroid cream to the eyelid."
- B. "Stop the antibiotics once symptoms improve."
- C. "Call your provider if your child's eye becomes difficult to open or moves painfully."
- D. "Keep the child's eye closed for 24 hours."

Worsening symptoms may mean progression to orbital cellulitis and need urgent care.



QUICK NCLEX REVIEW TABLE – PERIORBITAL VS. ORBITAL CELLULITIS

Feature	Periorbital Cellulitis	Orbital Cellulitis
Location	In front of orbital septum	Behind orbital septum
Pain with eye movement	✗ Absent	✓ Present
Vision changes	✗ None	✓ Possible
Proptosis (bulging eye)	✗ No	✓ Yes
Urgency	Urgent, but outpatient possible	Emergency – needs IV antibiotics/hospital
Common cause	Skin trauma, sinusitis	Sinusitis, progression from periorbital cellulitis



FINAL NCLEX TIPS FOR PERIORBITAL CELLULITIS

- Most common in children under 10
- Differentiate from orbital cellulitis by checking eye movement pain and vision
- Prompt antibiotic treatment is key
- Always educate caregivers on red flag symptoms: eye pain, vision loss, fever
- CT scan may be ordered to rule out deeper infection